

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning **(BL 5)**

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

Industry Problem Solving

Code of Conduct:

I K.ASWINI(192312397) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A retail company wants to recommend products to customers based on past purchases. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity is measured, how recommendations are generated, and how the system adapts to new customer data.
2. A manufacturing industry needs to predict machine output based on sensor readings that vary over time. Propose a solution using Locally Weighted Regression. Explain how local models improve prediction accuracy and how the system handles noisy or incomplete data.
3. A healthcare company aims to classify patients into risk categories using historical medical data. Suggest a model using KNN or RBF networks. Explain how the algorithm handles large datasets, feature selection, and accuracy improvement in real-world scenarios.
4. An e-commerce platform wants to detect similar products for dynamic pricing strategies. Design a solution using Case-Based Learning. Discuss how similarity metrics are defined, how past cases are reused, and how the system scales with increasing product data.
5. A logistics company needs to forecast delivery times based on location and traffic patterns. Propose a solution using Radial Basis Function networks or Locally Weighted Regression. Explain how the model captures non-linear relationships and improves prediction reliability under varying conditions.
6. A banking system wants to detect potential loan defaulters based on customer financial history and transaction patterns. Design a solution using K-Nearest Neighbour or Logistic Regression. Explain how similarity or decision boundaries are used, how features are selected, and how the system adapts to new customer data over time.
7. A smart agriculture system aims to predict crop yield based on soil quality, weather conditions, and irrigation levels. Propose a solution using Locally Weighted Regression or RBF networks. Explain how local learning improves predictions and how the system handles seasonal variations and noisy environmental data.
8. A cybersecurity firm wants to identify abnormal network behavior to detect intrusions. Design a system using KNN or Case-Based Learning. Explain how similarity between network patterns is measured, how anomalies are detected, and how the model updates with new attack patterns.
9. A streaming platform aims to recommend movies to users based on viewing history and ratings. Propose a solution using **KNN or Case-Based Learning**. Explain how user similarity is calculated, how recommendations are generated, and how the system handles large-scale user data efficiently.

10. A transportation company wants to optimize route selection based on traffic, distance, and fuel consumption. Design a solution using **RBF networks or Locally Weighted Regression**. Explain how the model captures complex relationships between variables and improves decision-making in dynamic and real-time environments.
11. A telecommunications company wants to recommend suitable data plans to customers based on their usage patterns. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity between users is measured, how recommendations are generated, and how the system adapts to changing user behavior over time.
12. An energy management company aims to predict electricity consumption based on factors such as time, weather, and appliance usage. Propose a solution using Locally Weighted Regression or RBF networks. Explain how local learning captures variations in consumption patterns and how the model handles noisy and seasonal data.
13. A fraud detection system in an online payment platform needs to classify transactions as legitimate or fraudulent. Design a solution using KNN or Logistic Regression. Explain how similarity or decision boundaries are used, how features are selected, and how the model adapts to evolving fraud patterns.
14. A manufacturing company wants to detect defective products based on sensor readings and quality measurements. Propose a solution using RBF networks or Case-Based Learning. Explain how non-linear relationships are captured, how similarity between cases is measured, and how the system improves with additional production data.
15. A smart city system aims to predict traffic flow and congestion levels using data from sensors, weather, and time of day. Design a solution using Locally Weighted Regression or KNN. Explain how the model captures dynamic patterns, how predictions are made in real-time, and how scalability is maintained with large volumes of data.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

ANSWERS

1.Problem:

The retail company has customer purchase history but does not know which products should be recommended to a particular customer.

Solution:

Use **K-Nearest Neighbour (KNN)**. Represent every customer using features such as number of purchases, product categories, spending amount, and ratings.

Problem Solving:

Suppose Customer A has purchased shoes, shirts, and watches. The system compares Customer A with all other customers using similarity. Assume the 3 nearest customers have also purchased sunglasses and bags. Since these products are not purchased by Customer A, the system recommends sunglasses and bags. Similarity can be calculated using Euclidean distance or cosine similarity. Smaller distance means greater similarity.

Result:

The system provides personalized product recommendations based on customers with similar purchasing behaviour.

2.Problem:

Machine output changes according to temperature, pressure, vibration, and speed. A single global regression model may not accurately represent all operating conditions.

Solution:

Use **Locally Weighted Regression (LWR)**.

Problem Solving:

Suppose the current machine condition is:

- Temperature = 70°C
- Pressure = 50 units
- Vibration = 10 units

The system searches historical sensor readings and gives **higher weights to observations close to these values**. Readings far away receive lower weights.

Result:

LWR provides accurate predictions because it focuses on machine conditions that are similar to the current condition.

3.Problem:

A healthcare company needs to classify patients as **Low Risk, Medium Risk, or High Risk**.

Solution:

Use **KNN** with features such as age, blood pressure, glucose level, cholesterol, and medical history.

Problem Solving:

Suppose a new patient has:

- Age = 60
- Blood Pressure = High
- Glucose = High
- Cholesterol = High

The system compares this patient with historical patients. Assume the 5 nearest patients are:

3 High Risk + 1 Medium Risk + 1 High Risk

Since most nearest patients are High Risk, the new patient is classified as **High Risk**.

Feature Selection:

Irrelevant features are removed and important medical features are normalized before calculating distance.

Large Dataset:

Efficient nearest-neighbour search and indexing can reduce the time required to find similar patients.

Result:

The patient is classified according to the risk category of the most similar historical patients.

4.Problem:

An e-commerce company wants to identify products similar to a new product so that it can determine suitable pricing.

Solution:

Use **Case-Based Learning**.

Problem Solving:

Suppose a new smartphone has:

- 8 GB RAM
- 128 GB storage
- Similar processor
- Similar camera
- Similar display

Similarity:

Similarity is calculated using product specifications, category, brand, price, and ratings.

Adaptation:

When new products and sales results are added, they become new cases for future decisions.

Result:

The system reuses previous product knowledge to identify similar products and support dynamic pricing.

5.Problem:

Delivery time depends on several factors such as distance, traffic, weather, and location. These relationships are non-linear.

Solution:

Use an **RBF Network**.

Problem Solving:

Suppose a delivery has:

- Distance = 15 km
- Heavy traffic
- Rainy weather

- Urban location

The RBF network compares this combination with previously learned delivery conditions.

If similar historical deliveries required approximately 55–65 minutes, the network predicts a delivery time within this range.

The hidden RBF layer calculates how close the current conditions are to learned patterns.

Result:

The system can predict delivery time more accurately because it learns complex relationships between traffic, distance, weather, and delivery time.

6.Problem:

A bank wants to identify customers who are likely to default on loans.

Solution:

Use **Logistic Regression**.

Problem Solving:

Consider a customer with:

- Low income
- Low credit score
- High existing debt
- Poor repayment history
- Large loan amount

These features are given to the logistic regression model.

Suppose the model produces:

Probability of default = 0.82

If the bank uses 0.5 as the decision threshold:

$0.82 > 0.5 \rightarrow \text{Potential Defaulter}$

If another customer has a probability of 0.20:

$0.20 < 0.5 \rightarrow \text{Low Default Risk}$

Result:

The bank can identify high-risk customers before approving loans.

7.Problem:

Crop yield depends on soil quality, rainfall, temperature, fertilizer, and irrigation. These factors vary between seasons.

Solution:

Use **Locally Weighted Regression**.

Problem Solving:

Suppose the current field has:

- Good soil
- Low rainfall
- High temperature
- Medium irrigation

The system searches historical agricultural records with similar conditions. These nearby observations receive higher weights.

If similar historical conditions produced:

Result:

The model provides a prediction based mainly on agricultural conditions similar to the current season.

8.Problem:

A cybersecurity company needs to identify whether network traffic is normal or an attack.

Solution:

Use **KNN**.

Problem Solving:

Suppose a new network connection has:

- Very high packet rate
- Long connection duration
- Large number of requests
- Unusual destination

The system compares this connection with historical network patterns.

Suppose the 5 nearest patterns are:

4 known attacks + 1 normal connection

Result:

The system can detect attacks by comparing current network behaviour with previously observed patterns.

9.Problem:

A streaming platform needs to recommend movies that a user is likely to watch.

Solution:

Use **KNN** to identify users with similar movie preferences.

Problem Solving:

Suppose User A has rated:

- Movie X $\rightarrow$ 5
- Movie Y $\rightarrow$ 4

Similarity:

Cosine similarity can be used to compare user rating vectors.

Scalability:

Sparse matrices and efficient nearest-neighbour search can handle large numbers of users.

Result:

The platform recommends movies based on the preferences of similar users.

10.Problem:

A transportation company must select the best route considering traffic, distance, and fuel consumption.

Solution:

Use an **RBF Network**.

Problem Solving:

Suppose there are three routes:

Route	Distance	Traffic	Fuel
A	10 km	Heavy	High
B	14 km	Low	Medium
C	12 km	Medium	Low

The RBF model evaluates the combined effect of these factors and predicts travel time and fuel cost.

Suppose the predicted total costs are:

- Route A $\rightarrow$ 80
- Route B $\rightarrow$ 55
- Route C $\rightarrow$ 65

Therefore, the system selects **Route B**.

Result:

RBF helps choose a route by learning complex relationships between distance, traffic, fuel consumption, and travel time.

11.Problem:

A telecom company wants to recommend the most suitable data plan to each customer.

Solution:

Use **KNN**.

Problem Solving:

Suppose Customer A uses:

- 80 GB data/month
- 500 minutes calls
- Low SMS usage

Adaptation:

If Customer A later starts using 150 GB per month, the customer profile is updated and a higher data plan can be recommended.

Result:

The system provides personalized plans based on current customer usage.

12.Problem:Electricity consumption changes with time, weather, season, and appliance usage.

Solution:

Use **Locally Weighted Regression**.

Problem Solving:

Suppose the current conditions are:

- Summer
- 38°C
- 2 PM
- High air-conditioner usage

The system searches historical observations with similar conditions and gives them higher weights.

Noise Handling:

Abnormal electricity readings can be detected and given lower weight. Seasonal data can be separately represented through local observations.

Result:

The system predicts electricity consumption according to similar time, weather, and usage conditions.

13.Problem:An online payment system must determine whether each transaction is legitimate or fraudulent.

Solution:

Use **Logistic Regression**.

Problem Solving:

Suppose a transaction has:

- Very high amount
- Unusual location
- New device
- Multiple transactions within a short time

The model processes these features and produces:

Fraud probability = 0.91

Using a threshold of 0.5:

$0.91 > 0.5 \rightarrow$ Fraudulent transaction

A transaction with probability 0.10 would be classified as legitimate.

Result:

The system identifies suspicious transactions using a learned probability-based decision boundary.

14.Problem:

A manufacturing company wants to detect defective products from sensor and quality measurements.

Solution:

Use an **RBF Network**.

Problem Solving:

Suppose a product has:

- High temperature
- High vibration
- Incorrect dimension
- Abnormal pressure

The RBF network compares this combination with learned defect patterns.

Suppose the current measurements are very close to a known defect pattern. The RBF neurons produce high activation, and the output class becomes:

Result:

The RBF network detects defects even when the relationship between sensor measurements and defects is non-linear.

15.Problem:

A smart city needs to predict traffic congestion using sensor readings, weather, location, and time.

Solution:

Use **KNN**.

Problem Solving:

Suppose the current situation is:

- Monday
- 8:30 AM
- Main Road
- Rainy weather
- High vehicle count

The system searches historical traffic records with similar conditions.

Suppose the 5 nearest records show:

High, High, Medium, High, High

Therefore:

4 out of 5 → High congestion

The system predicts:

HIGH TRAFFIC CONGESTION

Real-Time Processing:

New sensor readings are continuously supplied to the system. Efficient nearest-neighbour search can be used for large traffic datasets.

Result:

The city can predict congestion and take actions such as changing traffic signals or suggesting alternate routes.
